Adil Sukumar

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EDUCATION

Vellore Institute of Technology **Aug'25–May'30**
Integrated M.Tech. Artificial Intelligence & Bioinformatics | CGPA: 9.10/10

EXPERIENCE

British Airways **Oct'26–Present**
Data Science Job Simulation
- Explore customer-booking data and predictive analytics in an ongoing data science job simulation.

Pawsitive Diagnostics **Feb'26–Present**
Founder and CEO
- Lead **software, hardware and machine learning** development for a **wearable animal-health system** integrating **motion, acoustic, environmental and surface-temperature sensing**.
- Maintain **KiCad** schematics, **PCB layout**s and **Gerber** exports for the collar electronics; integrate device telemetry with edge-to-cloud monitoring.
- Develop **signal filtering**, **activity recognition** and context-aware alerts against individual baselines.

ElectraWireless **May'26–Aug'26**
Lead Intern, IoT & Embedded Systems
- Co-created **ELLY**'s **offline-first React/TypeScript** ecosystem with **Web-LLM** voice control, **TensorFlow.js/MobileNet** vision and **BLE/Wi-Fi** discovery.
- Automated **Capacitor** iOS and Android builds with **GitHub Actions**.
- Integrated BLE, Wi-Fi discovery, **MQTT** and WebSockets for local control and energy telemetry.

PROJECTS

Portiva | Java, Python, C++, Android, Arduino **2026**
- Built **mission state machines**, heading-aware **A* planning** and swept-body **collision checks** for a person-following delivery robot prototype.
- Implemented **Uno protocol**, **watchdog** and sensor-firmware workflows with an Android bench demo; kept motor outputs disabled pending physical commissioning.

SoftSpace and SoftFlex | Fusion, CAD, Python, FEA **2026**
- Developed **six adjustable headphone CAD variants** and a glasses-aware cushion concept, retaining editable **Fusion** history, **STEP** assemblies and analysis evidence.
- Audited **eight higher-order joint reference cases** and mesh-displacement comparisons; documented preload, friction and material assumptions plus required prototype tests.

LightSense Connected Lighting | ESP32, WS2812B, MQTT **2026**
- Built **ESP32** firmware with **FastAPI/SQLite** workflows for independent RGB and health-status LED installations.
- Published retained **MQTT** commands for reconnect recovery; configured the **MQTT** broker with **Docker Compose**.

Bluetooth Home Automation | Arduino, Relays, Bluetooth **2022**
- Built mobile-controlled switching for lights, fans and household appliances using Arduino, **relay modules** and Bluetooth communication.

TECHNICAL SKILLS

Firmware and Platforms: C/C++, Python, Arduino Uno, ESP32/ESP32-C3, Android
PCB and Electronics: KiCad, schematic capture, PCB layout, Gerber exports, relay interfacing
Connectivity: BLE/GATT, Bluetooth Classic, Wi-Fi, MQTT, WebSockets, serial protocols
Sensors and Telemetry: Sensor integration, JSON telemetry, filtering, individual baselines, anomaly alerts
Robotics and Control: Mission state machines, heading-aware A*, collision checking, watchdog/fault handling
CAD and Analysis: Autodesk Fusion, STEP assemblies, FEA, mesh-displacement comparison, load-case analysis
Edge AI: TensorFlow.js, MobileNet, Web-LLM, ONNX, activity-recognition workflows
Engineering Tools: Git, Linux, Docker Compose, FastAPI, SQLite, GitHub Actions, Capacitor

LEADERSHIP AND ACHIEVEMENTS

VIT Student Council — Member Secretary – Technical | Sep'26–Present
WiCyS Student Chapter, VIT — Secretary | Dec'25–Present
- CyberShield Hackathon 2025, VIT — **First Place**.
- SKITECH National Innothon 3.0, Indore — **First Runner-Up**, national level.
- SASI National Hackathon 2026, Andhra Pradesh — **Top 6**, national level.